

CARE-PROGRESS India

Packet-Collected Dataset Protocol for Outpatient Care-Loop Progression

Abnormal Diagnostic Results | Referrals | Follow-Up Loops | Risk-Weighted Delay

Core thesis

CARE-PROGRESS India studies time-sensitive outpatient care progression as a temporal workflow problem. Using packet-collected audits of abnormal diagnostic results and referral/follow-up loops, the dataset captures clinical urgency, event sequences, handoffs, workflow ownership, tracking systems, patient-contact pathways, closure status, and risk-weighted delay. The resulting benchmark supports regression, survival analysis, temporal sequence modeling, dynamic workflow graph learning, stall-stage prediction, and observational counterfactual analysis of modifiable care-delivery structures.

Protocol snapshot

Document purpose	Complete field protocol for building the survey website, printed handout, data dictionary, collection workflow, quality checks, and early analysis plan.
Collection mode	Remote packet or survey website. No in-person experiments required.
Target collection window	Two-week pilot, with a one-week field push possible if the website is ready.
Minimum serious unit	One specific recent care loop with clinical urgency, event timing offsets, workflow path, resolution status, evidence source, and confidence.
Do not collect	Patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, raw notes, WhatsApp screenshots, prescriptions, lab-report photos, or any identifiable information.

Version 1.0 - Field-ready, survey-website specification

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1. Executive Summary

CARE-PROGRESS India is a packet-collected, multi-site pilot dataset for modeling how time-sensitive outpatient care loops progress through real clinical workflows. It is designed for fast collection through a survey website or printed QR packet while preserving enough structure for credible medical operations research and machine-learning benchmarking.

The dataset deliberately avoids a generic 'patient readiness' or 'clinic workflow opinion' frame. Instead, it uses specific recent care-loop audits. Each row describes one abnormal diagnostic result or referral/follow-up loop, including clinical urgency, recommended closure window, event timing offsets, actor-role pathway, handoff count, tracking method, closure status, evidence source, and confidence.

Core object

The scientific object is not delay itself. Delay is the symptom. The scientific object is care progression under clinical time-sensitivity and uncertainty.

1.1 What the dataset should be able to support

- Regression: days to review, days to contact, days to action, days to closure, days overdue, risk-weighted delay.
- Survival analysis: closure probability over time, with censored still-open loops.
- Temporal sequence modeling: next event, time to next event, final closure status.
- Dynamic workflow graph learning: role-to-role paths such as Lab -> Physician -> Coordinator -> Patient -> Specialist -> Closure.
- Stall-stage prediction: whether the loop stalls before review, before patient contact, before documented action, before closure, or during referral completion.
- Observational counterfactual analysis: whether modifiable workflow structures, such as clear ownership or staff tracking, are associated with faster closure after accounting for urgency and loop type.

2. Why CARE-PROGRESS Is Not Just a Delay Audit

A simple delay audit asks, 'How long did it take?' That is not enough. Everyone already knows delays exist. A Mayo-style health-systems researcher will ask whether the delay was clinically meaningful. A NeurIPS-style reviewer will ask whether the dataset creates a non-trivial learning problem.

CARE-PROGRESS answers both concerns by connecting workflow progression to clinical time-sensitivity. A 14-day delay is not inherently good or bad. It depends on whether the loop was a routine referral, an abnormal HbA1c, suspicious mammogram follow-up, abnormal pathology, post-procedure review, or urgent oncology referral. The dataset therefore captures both actual delay and recommended closure window, allowing risk-weighted delay to be derived.

Weak framing	Strong framing
Measure clinic delays	Model care progression under clinical urgency
Predict whether follow-up happened	Predict where a time-sensitive care loop stalls and when it

	closes
Generic workflow survey	Specific recent loop audit with evidence and confidence
Binary outcome only	Regression, survival, sequence, graph, stall-stage, and counterfactual tasks
Delay as endpoint	Risk-weighted delay and resolution status as clinically interpretable outcomes
Non-trivial insight target The dataset should reveal which operational structures explain delay after accounting for clinical urgency and loop type. Examples: clear owner vs no clear owner, staff-tracked vs memory-tracked, EHR task vs paper/WhatsApp, low handoff vs high handoff, support staff involved vs not involved.	

3. Dataset Scope and Scientific Object

3.1 Collect only two care-loop families first

The fastest route to a defensible two-week dataset is narrow scope. Do not collect every workflow problem. The initial dataset should focus on two families that are clinically meaningful, common, time-sensitive, sequential, record-backed, and vulnerable to handoff failure.

Family	Examples	Why it works
A. Abnormal diagnostic result loops	Abnormal lab, abnormal imaging, abnormal pathology, abnormal mammogram, abnormal HbA1c, abnormal creatinine/eGFR, suspicious imaging, biopsy/pathology follow-up	Clear index event, natural review/contact/action/closure sequence, clinically time-sensitive, often documented in logs or charts.
B. Referral / follow-up completion loops	Specialist referral, oncology referral, cardiology referral, imaging referral, procedure referral, post-procedure follow-up, chronic follow-up, repeat test order	Closure is not guaranteed, handoffs are common, loss-to-follow-up is measurable, and workflow architecture matters.

3.2 What not to collect in the pilot

- General doctor workflow opinions as the primary dataset.
- Generic patient readiness scores.
- Medication refill workflows unless a clinic specifically has clean records; these can be added later.
- Broad qualitative essays about clinic problems.
- Any row that refers to what 'usually' happens rather than one specific recent care loop.

3.3 Definition of a care loop

Care loop definition

A care loop is a clinical task that begins at a specific index event and should later be reviewed, acted upon, escalated, closed, or explicitly marked as no action needed. Each row in CARE-PROGRESS is one

care loop.

4. Collection Strategy and Two-Week Targets

The goal is maximum useful signal per minute of respondent effort. A perfectly rigorous hospital extraction that takes months is not viable. A 2,000-row opinion survey collected in a day is also not useful. CARE-PROGRESS uses a layered compromise: many specific loop audits, evidence grading, confidence scoring, and a small optional validation subset.

4.1 Respondents

- Physicians: strongest for urgency and clinical consequence.
- Nurses / medical assistants: often know patient contact and follow-up status.
- Clinic coordinators / receptionists: often know referral scheduling and contact attempts.
- Lab/referral coordinators: often know result availability and referral closure.
- Administrators: useful for site operations and future data export feasibility, but not sufficient for loop-level labels alone.

4.2 Minimum respondent task

Per respondent requirement

Each respondent audits 6 specific recent care loops: ideally 3 abnormal diagnostic result loops and 3 referral/follow-up loops. Additional loops up to 10 are optional.

Target	Respondents	Loops per respondent	Total loop rows	Interpretation
Conservative	30	6	180	Viable seed pilot if evidence quality is good.
Good	50	6	300	Strong two-week pilot for preliminary analysis.
Excellent	100	6	600	Very strong for remote packet collection.
Stretch	100	10	1,000	Impressive only if not shallow or memory-only.

4.3 Selection rule

Use the most recent consecutive eligible loops from the last 30-90 days. Do not choose only memorable, severe, successful, or failed cases. This reduces cherry-picking and improves interpretability. If a respondent intentionally selects high-risk examples only, the row may still be useful for modeling but should not be used to estimate prevalence.

5. Survey Website Architecture

Build the website as a structured data-collection instrument, not a generic survey. The repeated object is the care loop. The website should produce clean CSV exports, enforce de-identification, validate timeline fields, and minimize free text.

5.1 Required website flow

1. Landing page: study purpose, time required, what counts as a care loop, de-identification warning.
2. Consent gate: mandatory checkbox confirming no patient identifiers will be entered.
3. Respondent/site context: 2-3 minute context module.
4. Loop audit dashboard: show 6 required loops, with progress indicator and optional extra loops.
5. Repeated care-loop form: one structured form per loop.
6. Operational structure module: named responsibility, unresolved-loop review process, future data export feasibility.
7. Final verification page: confirmation, optional willingness for de-identified verification, and submit.

5.2 UX rules

- Mobile-first. Many respondents will use phones.
- Total completion target: 13-19 minutes. Hard cap: 25 minutes.
- Six loops required; more loops optional. Do not require 10 by default.
- Use dropdowns and radio buttons. Avoid long free text.
- Show progress: Context -> Loop 1 -> Loop 2 -> ... -> Submit.
- Autosave if possible. If not, keep pages short.
- Put a patient-identifier warning above every optional free-text box.
- Provide examples for day offsets: same day = 0, three days later = 3.

5.3 Validation rules

Rule	Purpose
No submission without de-identification confirmation	Prevents accidental patient-identifiable data entry.
Every loop must have loop family, loop type, urgency, recommended window, status, evidence source, and confidence	Ensures the row can be analyzed.
Day offsets must be numeric when status is known	Keeps regression targets usable.
Closure day cannot be earlier than review/action day if both are known	Prevents impossible timelines.
Evidence source and confidence required	Separates record-backed rows from memory-backed rows.
Selection method required	Distinguishes consecutive sampling from convenience or high-risk-only sampling.

5.4 Recommended tech stack

- Fast custom build: Next.js or React frontend, Supabase backend, Vercel deployment, CSV export from Supabase.

- No-code fallback: Tally, Typeform, Google Forms, or KoboToolbox. Use only if custom build would consume field time.
- Do not build login, dashboards, analytics, AI models, or hospital account systems for the pilot. The goal is clean data capture.

6. Printed Packet and QR Handout

Print a short field packet as a trust and backup layer, but make the website the primary data-capture mechanism. The printed packet should not be the full research master document. It should be short enough to hand to busy doctors and staff without triggering immediate administrative despair.

6.1 Four-page field packet

Page	Content	Purpose
1	Study intro, purpose, time required, QR code, no patient identifiers warning	Legitimacy and clarity.
2	Definitions: care loop, two loop families, selection rule, day-offset examples	Prevents confusion and cherry-picking.
3	Six-row care-loop audit table	Allows offline completion or quick preview before using website.
4	Short respondent/site context and operational readiness	Captures workflow metadata and future data feasibility.
Field instruction The QR form is preferred because it exports directly into the dataset. The paper version is only for situations where online access is inconvenient.		

7. Full Survey Instrument: Website Order

This section is the full survey in the recommended order. It can be copied into a custom website, Tally, Typeform, Google Forms, REDCap, KoboToolbox, or another structured form tool. Required fields are marked REQUIRED. Optional fields may be hidden unless the respondent selects a relevant answer.

7.1 Page 1 - Landing Page Text

Copy-paste landing page

CARE-PROGRESS India studies how time-sensitive outpatient care loops move from event creation to review, patient contact, action, escalation, and closure. The goal is to understand which workflow structures predict clinically meaningful delay, stalling, escalation, or loss-to-follow-up.

You will be asked to audit 6 recent care loops from the last 30-90 days. A care loop may be an abnormal diagnostic result, referral, follow-up request, repeat test order, post-procedure follow-up, or patient-contact task. Use records where possible. Do not enter any patient-identifying information.

Estimated time: 13-19 minutes.

7.2 Page 2 - Consent and De-identification Gate

Question	Response options	Required
I confirm that I will not enter patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, prescription images, lab-report photos, WhatsApp screenshots, raw clinical notes, or other identifiable information.	☐ I confirm	Yes
I understand this is a de-identified research audit of outpatient care-loop progression, not a doctor performance evaluation and not a clinical decision-making tool.	☐ I confirm	Yes
Are you using clinic records where possible?	Mostly records / Mixed records and memory / Mostly memory	Yes

7.3 Page 3 - Respondent and Site Context

Question	Response options	Required
Respondent role	Physician / Nurse / Medical assistant / Clinic coordinator / Receptionist / Lab staff / Admin / Other	Yes
Specialty or service area	General medicine / Cardiology / Endocrinology / Oncology / OB-GYN / Pediatrics / Surgery / Orthopedics / Neurology / Multi-specialty / Other	Yes
Practice setting	Solo private clinic / Group private clinic / Private hospital OPD / Government hospital OPD / Charitable clinic / Telemedicine / Other	Yes
Average outpatient volume per day	<20 / 20-39 / 40-59 / 60-79 / >=80 / Unknown	Yes
Support staff usually available	Receptionist / Nurse / MA / Coordinator / Lab coordinator / Pharmacist / None / Other	Yes
Main record system	Paper / EHR / Clinic software / WhatsApp/photos / Mixed / Other	Yes
Lab/result tracking method	None / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes

Referral tracking method	None / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes
Follow-up tracking method	None / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes
Main patient communication channel	Phone / WhatsApp / SMS / In-person / Portal/app / Email / Other	Yes
Digital maturity score	1 mostly paper / 2 paper + phone / 3 spreadsheet or WhatsApp tracking / 4 clinic software / 5 integrated EHR or task system	Yes

8. Care-Loop Audit Questions

This repeated module is the core of the dataset. Each respondent should complete it 6 times: 3 abnormal diagnostic result loops and 3 referral/follow-up loops. Each loop should be a specific recent loop from the last 30-90 days, preferably selected using the most recent consecutive eligible loops.

8.1 Loop Identity

Question	Response options	Required
Loop code	L001, L002, L003, etc. Do not use patient identifiers.	Yes
Loop family	Abnormal diagnostic result / Referral or follow-up	Yes
Specific loop type	Abnormal lab / Abnormal imaging / Abnormal pathology / Specialist referral / Imaging referral / Procedure referral / Follow-up appointment / Repeat test / Post-procedure follow-up / Other	Yes
Specialty involved	General medicine / Cardiology / Endocrinology / Oncology / OB-GYN / Pediatrics / Surgery / Orthopedics / Neurology / Other	Yes
For abnormal result loops: test category	CBC / HbA1c / renal function / liver function / thyroid / lipid / X-ray / ultrasound / CT/MRI / mammogram / pathology / other / N/A	Conditional
For referral/follow-up loops: referral or follow-up category	Cardiology / oncology / endocrinology / imaging / surgery / OB-GYN / pediatrics / post-procedure / chronic follow-up / other / N/A	Conditional

8.2 Clinical Time-Sensitivity

Question	Response options	Required
Clinical urgency	Routine / Semi-urgent / Urgent / Critical / Unclear	Yes
Clinical severity	Mild / Moderate / Severe / Critical / Not applicable / Unclear	Yes
Recommended closure window	Same day / 1-3 days / 4-7 days / 8-14 days / 15-30 days / >30 days / Unclear	Yes
Could delay plausibly affect care?	No / Possibly / Yes / Unclear	Yes

Why this matters

Raw delay is not enough. The same 14-day delay may be acceptable for a routine referral and dangerous for a severe abnormal result. Urgency and recommended closure window allow risk-weighted delay to be derived.

8.3 Event Timeline

Use relative day offsets, not actual dates. The index event is Day 0. If an event happened the same day, enter 0. If it happened three days later, enter 3. If the event did not happen, was not needed, or is unknown, use the status field rather than inventing a number.

Field	Response / entry	Required
Index event	Automatically defined as Day 0	Yes
Clinician review status	Reviewed / Not reviewed / Not needed / Unknown	Yes
Clinician review day offset	Numeric days after Day 0 if known	Conditional
Patient contact status	Contacted / Not contacted / Not needed / Unknown	Yes
Patient contact day offset	Numeric days after Day 0 if known	Conditional
Documented action status	Action documented / No action needed / Action not documented / Unknown	Yes
Documented action day offset	Numeric days after Day 0 if known	Conditional
Follow-up scheduled status	Scheduled / Not scheduled / Not needed / Unknown	Yes
Follow-up scheduled day offset	Numeric days after Day 0 if known	Conditional
Loop closure status	Closed / Still open / Lost to follow-up / Escalated / Closed as no action needed / Unknown	Yes
Loop closure day offset	Numeric days after Day 0 if closed	Conditional

If still open: audit age in days	How many days old was the loop at audit? Numeric days if known.	Conditional
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8.4 Workflow Path / Graph Fields

Question	Response options	Required
Who first received or noticed the loop?	System alert / Physician / Nurse / Receptionist / Coordinator / Lab staff / Patient / Unknown	Yes
Who reviewed it clinically?	Physician / Nurse / Specialist / Not reviewed / Unknown	Yes
Who contacted the patient?	Physician / Nurse / Receptionist / Coordinator / Patient not contacted / Not needed / Unknown	Yes
Who primarily owned closure?	Physician / Nurse / Coordinator / Receptionist / Patient / No clear owner / Unknown / Other	Yes
Who closed the loop?	Physician / Nurse / Coordinator / Receptionist / System / Still open / Unknown	Yes
Number of handoffs	0 / 1 / 2 / 3 / 4+ / Unknown	Yes

These role fields allow graph paths to be reconstructed without collecting personal names. Example: Lab -> Physician -> Receptionist -> Patient -> Specialist -> Closure.

8.5 Workflow Structure

Question	Response options	Required
Tracking method for this loop	No formal tracking / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes
Support staff involved?	Yes / No / Unclear	Yes
Main communication channel	Phone / WhatsApp / SMS / In-person / Portal/app / Other / Unknown	Yes
Approximate outpatient load that day	<20 / 20-39 / 40-59 / 60-79 / >=80 / Unknown	Optional but recommended

8.6 Resolution and Clinical Consequence Proxy

Question	Response options	Required
Current status	Resolved / Still open / Lost to follow-up / Escalated / Closed as no action needed / Unknown	Yes

Was the patient successfully contacted?	Yes / No / Not needed / Unknown	Yes
Was follow-up completed?	Yes / No / Not needed / Unknown	Yes
Was urgent escalation required?	Yes / No / Unclear	Yes
Was emergency referral/admission documented?	Yes / No / Unclear	Yes
Was the care plan changed because of this loop?	Yes / No / Unclear	Yes

Do not overclaim patient harm from this packet. Use these as clinical consequence proxies unless independently adjudicated medical outcomes are later collected.

8.7 Evidence Quality

Question	Response options	Required
Evidence source	EHR / Paper chart / Lab log / Referral register / Appointment register / Staff call log / Physician memory / Staff memory / Other	Yes
Confidence	Low / Medium / High	Yes
Selection method	Most recent consecutive loops / Randomly selected loops / Convenience sample / High-risk examples only / Unknown	Yes
May a de-identified subset be verified later by an abstractor?	Yes / No / Maybe	Optional

9. Data Schema and Derived Variables

The survey website can store one large table at first, but the export should be structured as if the following relational tables exist. This preserves compatibility with later graph and sequence modeling.

9.1 Export tables

Table	Main contents
respondents.csv	Respondent role, specialty, practice setting, outpatient volume, record systems, tracking systems, communication channels, digital maturity.
careLoops.csv	One row per loop: loop family, type, urgency, severity, recommended window, closure status, evidence quality.
eventTimeline.csv	Review, contact, action, follow-up scheduled, closure statuses and day offsets.
workflowPath.csv	First receiver, reviewer, contact role, owner role, closer role,

	handoff count.
resolutionOutcomes.csv	Resolved/still open/lost/escalated/no action needed, contact success, follow-up completion, escalation, care-plan change.
evidenceQuality.csv	Evidence source, confidence, selection method, verification willingness.
siteOperations.csv	Named owners, unresolved-loop review cadence, future data export feasibility, approval pathway.

9.2 Core columns

loop_id, respondent_id, respondent_role, specialty, practice_setting, loop_family, loop_type, urgency, severity, recommended_window_days, review_status, review_day_offset, contact_status, contact_day_offset, action_status, action_day_offset, followup_scheduled_status, followup_scheduled_day_offset, closure_status, closure_day_offset, audit_age_days_if_open, first_receiver_role, reviewer_role, contact_role, owner_role, closer_role, handoff_count, tracking_method, support_staff_involved, communication_channel, clinic_load, patient_contacted, followup_completed, escalation_required, emergency_referral_or_admission, care_plan_changed, evidence_source, confidence, selection_method

9.3 Derived variables

Derived variable	Definition
days_to_review	review_day_offset if review_status = Reviewed and offset is known.
days_to_contact	contact_day_offset if patient contact was needed and known.
days_to_action	action_day_offset if action was documented and known.
days_to_followup_scheduled	followup_scheduled_day_offset if scheduled and known.
days_to_closure	closure_day_offset if closure_status indicates closed/resolved/closed as no action needed and offset is known.
censored_status	1 if loop is still open at audit or closure time is unknown but event is not known to be closed; 0 if closure time known.
censored_time	audit_age_days_if_open for still-open loops.
recommended_window_days	Convert window category to representative value: same day=0, 1-3=3, 4-7=7, 8-14=14, 15-30=30, >30=60 unless locally specified.
days_overdue	days_to_closure - recommended_window_days, only when closure time is known.
positive_days_overdue	max(days_overdue, 0).
urgency_weight	Routine=1, Semi-urgent=2, Urgent=3, Critical=5, Unclear=missing.
risk_weighted_delay	positive_days_overdue * urgency_weight.

stall_stage	Stage where progression stopped or largest delay accumulated: pre-review, contact, action, scheduling, closure, referral completion, documentation unknown.
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10. Data Quality, Evidence Grading, and Validation

The dataset will contain rows of different epistemic strength. That is acceptable only if quality is explicitly graded. Do not launder memory-based answers into ground truth.

Grade	Definition
Grade A	Record-backed, high confidence, complete timeline, specific loop, selection method known.
Grade B	Record-backed or mixed evidence, medium/high confidence, partial timeline but usable closure or censoring information.
Grade C	Memory-based but tied to one specific loop, with usable urgency, status, and approximate timing.
Grade D	Vague, missing evidence source, no usable timeline, generic 'usually' answers, or possible patient identifiers. Drop from main analysis.

10.1 Validation subset

- Ask each respondent whether a de-identified subset may be verified later.
- If possible, independently verify 10-20% of loop rows at cooperative sites.
- Prioritize verification of high-risk, still-open, escalated, and lost-to-follow-up loops.
- Report agreement for closure status, timeline offsets, urgency category, and evidence source.
- Use verified rows as a gold subset; use the rest with evidence-grade sensitivity analysis.

10.2 Cleaning rules

- Remove or redact any row containing patient identifiers.
- Flag impossible timelines, such as closure before index event or contact before review if clinically impossible for that loop.
- Separate high-risk-example-only rows from consecutive/random rows.
- Do not use rows with unclear urgency for risk-weighted delay unless urgency can be coded later.
- Analyze Grade A/B rows first; add Grade C in sensitivity analysis only.

11. Analysis and Benchmark Tasks

The first analysis should not be a neural network. Begin with descriptive statistics, variance decomposition, delay distributions, survival curves, and interpretable regression. Then define ML benchmarks that beat obvious baselines.

11.1 Descriptive analysis

- Median and IQR of days_to_review, days_to_contact, days_to_action, days_to_closure.

- Closure curves by loop family and urgency.
- Stall-stage frequencies by loop type.
- Risk-weighted delay by tracking method, owner role, handoff count, and support staff involvement.
- Evidence-grade sensitivity: Grade A/B only versus all usable rows.
- Missingness by respondent role, specialty, loop family, and evidence source.

11.2 Benchmark tasks

Task	Target	Why it matters
Time-to-closure regression	days_to_closure	Measures how long a care loop takes to complete.
Risk-weighted delay regression	risk_weighted_delay	Combines delay with clinical time-sensitivity.
Survival modeling	P(closure by t), censored still-open loops	Naturally handles unresolved loops.
Stall-stage prediction	pre-review/contact/action/scheduling/closure/lost/unknown	Identifies where the workflow fails.
Next-event prediction	next workflow state and time to next event	Turns loop progression into temporal sequence modeling.
Workflow graph prediction	closure time, stall stage, or delay from role-path graph	Uses handoffs and actor roles rather than static features only.
Observational counterfactual analysis	effect of owner clarity, tracking system, handoff count	Tests modifiable workflow structures.

11.3 Baselines

- Urgency-only model.
- Loop-type-only model.
- Specialty-only model.
- Tracking-method-only model.
- Site/practice-setting-only model.
- Full workflow model using urgency, loop type, owner role, handoffs, tracking method, support staff, communication channel, and evidence source.

The full model must beat the obvious baselines meaningfully. If it does not, the dataset may still be useful as descriptive operations research, but the ML contribution is weaker.

11.4 Metrics

- Regression: MAE, RMSE, median absolute error, Spearman correlation, explained variance/R².
- Survival: C-index, integrated Brier score, calibration of closure probability by time window.
- Classification: macro-F1, balanced accuracy, AUROC/AUPRC for loss-to-follow-up or escalation if used as secondary targets.
- Calibration: predicted delay deciles versus observed delay.

12. Recruitment, Field Workflow, and Timeline

12.1 Two-week execution plan

Days	Actions	Success criteria
1-2	Build website, test fake entries, create QR packet, prepare recruitment message.	Functional form, CSV export, no identifier leakage in test entries.
3-4	Pilot with 3-5 doctors/staff. Collect 20-30 loops. Fix confusing questions.	Each loop takes under 2.5 minutes; day offsets understood.
5-11	Full collection push through clinics, doctor contacts, WhatsApp groups, coordinators, site champions.	At least 50 respondents or 300 loop rows if possible.
12-13	Clean, grade, de-identify, remove bad rows, flag evidence quality and selection method.	Grade A/B main dataset created; Grade C sensitivity set separated.
14	Preliminary analysis and field report.	Delay distributions, closure curves, stall-stage table, risk-weighted delay by workflow structure.

12.2 Recruitment script

Message to send

Dear Doctor / Clinic Team,

I am conducting a short research pilot on outpatient care-loop progression in Indian clinical settings. The study examines how abnormal diagnostic results and referral/follow-up tasks move from review to patient contact, action, escalation, and closure.

The survey takes about 13-19 minutes. It asks you to audit 6 recent care loops from the last 30-90 days using records where possible. Please do not enter patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, screenshots, prescriptions, lab-report photos, or other identifying information.

The purpose is research and dataset development, not doctor performance evaluation or commercial marketing.

Survey link: [insert link]

QR code: [insert QR]

Thank you for helping measure where outpatient care progression succeeds or stalls.

12.3 Follow-up reminders

- Reminder 1 after 24 hours: emphasize it takes under 20 minutes and only 6 loops are required.

- Reminder 2 after 72 hours: emphasize no patient identifiers and that staff/coordinators may complete record-backed loop audits.
- Final reminder: ask for at least 3 loops if 6 is too much; partial data is better than silence, but mark incomplete submissions.

13. Ethics, Privacy, and Safety Boundaries

CARE-PROGRESS should be treated as research data collection involving healthcare workflow information. Even when patient identifiers are not collected, clinics may have internal approval requirements. Obtain site permission or ethics guidance where required. This protocol is not legal advice and should be reviewed by appropriate institutional or ethics personnel before scaled use.

13.1 Hard privacy rules

- Do not collect patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, prescription images, raw notes, WhatsApp screenshots, lab-report photos, or identifiable dates tied to rare events.
- Use loop codes such as L001, not patient IDs.
- Use relative day offsets, not exact dates, whenever possible.
- Use role categories rather than staff names.
- Use broad specialties and categories rather than rare identifiable details.
- Remove or redact any accidental identifiers before analysis.

13.2 Claims boundary

- Do not claim measured patient harm unless independently verified clinical outcomes are collected later.
- Use phrases such as clinical consequence proxy, escalation, lost-to-follow-up, unresolved loop, delayed closure, or care plan changed.
- Do not use the dataset to judge individual doctors or clinics.
- Do not deploy a model for clinical decisions based on the pilot. This is research and feasibility data.

Appendix A - Copy-Paste Website Form

Use this condensed version to build the first website quickly. The full instrument above contains the rationale and detailed wording.

A1. Consent Gate

- Checkbox: I will not enter patient identifiers or identifiable clinical documents.
- Checkbox: I understand this is a de-identified research audit, not performance evaluation or clinical decision support.
- Question: Are you using clinic records where possible? Mostly records / Mixed / Mostly memory.

A2. Respondent Context

- Role, specialty, practice setting, outpatient volume, support staff, record system, lab tracking, referral tracking, follow-up tracking, communication channel, digital maturity.

A3. Loop Audit - Repeat 6 times

- Loop code, family, specific type, specialty, urgency, severity, recommended closure window, delay could affect care.
- Review status/day, contact status/day, action status/day, follow-up scheduled status/day, closure status/day, audit age if open.
- First receiver, reviewer, contact role, owner role, closer role, handoff count.
- Tracking method, support staff involved, communication channel, clinic load.
- Resolution status, patient contacted, follow-up completed, escalation required, emergency referral/admission, care plan changed.
- Evidence source, confidence, selection method, verification willingness.

Appendix B - Printed Four-Page Packet Layout

Page 1: Intro and QR

Title: CARE-PROGRESS India: Outpatient Care-Loop Progression Audit

- Purpose: understand how abnormal results and referral/follow-up loops move through review, contact, action, escalation, and closure.
- Time: 13-19 minutes.
- Task: audit 6 recent loops from last 30-90 days.
- Privacy: no patient identifiers or raw documents.
- QR code and short link.

Page 2: Definitions

- Care loop: a clinical task that begins at a specific event and should later be closed.
- Loop families: abnormal diagnostic result and referral/follow-up.
- Selection: most recent consecutive eligible loops; do not choose only memorable or failed cases.
- Day offset examples: same day = 0; three days later = 3.

Page 3: Six-Row Care-Loop Table

Columns: loop code, loop family, loop type, urgency, recommended window, review day, contact day, action day, closure day, status, handoffs, tracking method, owner, evidence source, confidence.

Page 4: Respondent/Site Context

Role, specialty, setting, outpatient volume, record system, tracking system, communication channel, digital maturity, future verification willingness.

Appendix C - Example Completed Row

Field	Example value
loop_id	L014
loop_family	Abnormal diagnostic result

loop_type	Abnormal HbA1c
urgency	Semi-urgent
recommended_window_days	7
review_day_offset	2
contact_day_offset	5
action_day_offset	5
closure_day_offset	14
owner_role	Physician + receptionist/coordinator
handoff_count	2
tracking_method	Paper log
resolution_status	Resolved late
evidence_source	Lab log + paper chart
confidence	High
Derived days_overdue	$14 - 7 = 7$
Derived risk_weighted_delay	$7 * 2 = 14$

Appendix D - Data Dictionary: Selected Fields

Field	Type	Meaning
loop_id	string	Unique loop code within respondent; never a patient identifier.
respondent_id	string	Generated anonymous respondent/session ID.
loop_family	categorical	Abnormal diagnostic result or Referral/follow-up.
loop_type	categorical	Specific subtype such as abnormal lab, abnormal imaging, specialist referral.
urgency	categorical	Routine, semi-urgent, urgent, critical, unclear.
recommended_window_days	numeric	Converted expected closure window in days.
review_day_offset	numeric	Days from Day 0 to clinical review, if known.
contact_day_offset	numeric	Days from Day 0 to patient contact, if

		known and needed.
action_day_offset	numeric	Days from Day 0 to documented action, if known.
closure_day_offset	numeric	Days from Day 0 to loop closure, if closed.
audit_age_days_if_open	numeric	Age of loop at audit if still open; used for censoring.
first_receiver_role	categorical	Role that first received/noticed the loop.
owner_role	categorical	Role primarily responsible for closure.
handoff_count	ordinal	0, 1, 2, 3, 4+, unknown.
tracking_method	categorical	Memory, paper, spreadsheet, software, EHR task, WhatsApp, mixed, none.
evidence_source	categorical	EHR, paper chart, lab log, referral register, appointment register, staff call log, memory, other.
confidence	categorical	Low, medium, high.

Appendix E - Quality Assurance Checklist

- Does every row refer to one specific recent loop, not a general pattern?
- Does every row have loop type, urgency, recommended window, closure status, evidence source, confidence, and selection method?
- Are there patient identifiers? If yes, remove/redact before analysis.
- Are timelines possible? Flag impossible offsets.
- Are still-open loops coded with audit age for censoring?
- Are high-risk-only samples separated from consecutive/random samples?
- Are memory-only rows separated from record-backed rows?
- Can Grade A/B rows be analyzed separately from Grade C rows?
- Can stall stage be derived from the available event statuses?
- Can risk-weighted delay be derived only when urgency and closure/recommended window are known?

Appendix F - What to Say When Handing the Packet

Thirty-second explanation

This is a short research audit on how outpatient care loops progress. We are not asking for patient names or documents. We only need 6 recent examples, such as abnormal results or referrals, and rough day offsets showing when they were reviewed, when the patient was contacted, when action happened, and whether the loop closed. The QR form is preferred because it exports cleanly; the paper packet is just a backup.

Final Operating Principle

The line between dataset and vibes census

Never ask what usually happens. Ask what happened in one specific recent care loop. Capture its urgency, timeline, workflow path, resolution, evidence, and confidence. Then repeat.